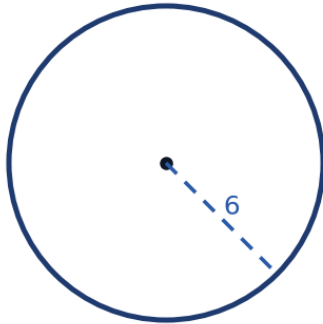


# Circles & Coordinate Geometry



## The equation of a circle

- 1 Write the equation of a circle with center  $(-4, 7)$  and radius 9.
- 2 The circle below has the radius shown. Find its area and circumference, in terms of  $\pi$ .



- 3 A circle has diameter 26 and is centered at the origin. Write its equation, and state its radius.

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## Completing the square

- 4 Find the center and radius of  $x^2 + y^2 - 6x + 8y - 11 = 0$ .
- 5 Find the center and radius of  $x^2 + y^2 + 10x - 4y + 13 = 0$ .

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## Distance, midpoint, and slope

- 6 Find the distance between  $(-3, 5)$  and  $(4, -7)$ , to the nearest hundredth.
- 7 Find the midpoint and slope of the segment joining  $(2, -6)$  and  $(-8, 10)$ .
- 8 Find the slope of the line through  $(-2, 7)$  and  $(6, -5)$ , and state whether the line rises or falls from left to right.

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## Coordinate proofs

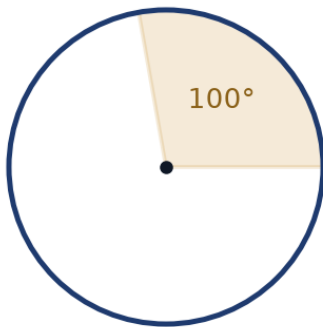
- 9 Show that the triangle with vertices  $A(0, 0)$ ,  $B(4, 0)$ ,  $C(4, 3)$  is a right triangle, using slopes.
- 10 Show that  $A(1, 2)$ ,  $B(4, 3)$ ,  $C(3, 6)$ ,  $D(0, 5)$  form a parallelogram, by showing both pairs of opposite sides are parallel.
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### Tangent lines to circles

- 11 A circle has center  $O(2, 3)$  and radius 5. Point  $P(2, 12)$  lies outside the circle. Find the length of the tangent from  $P$  to the circle, to the nearest hundredth.
- 12 Find the equation of the tangent line to the circle  $x^2 + y^2 = 25$  at the point  $(3, 4)$  on the circle.
- 13 A circle has center  $(0, 0)$  and radius 10. Determine whether the point  $(6, 9)$  lies inside, on, or outside the circle.
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### Circle theorems

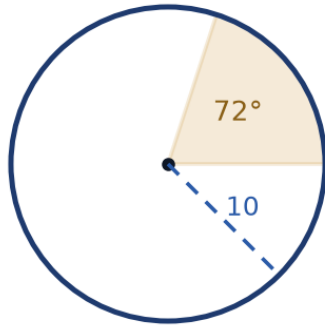
- 14 A central angle of  $84^\circ$  subtends an arc. Find the measure of the inscribed angle subtending the same arc.
- 15 A chord subtends the central angle shown below. Find the measure of the inscribed angle subtending the same (minor) arc.



### Arc length and sector area

- 16 A circle has radius 12 cm. Find the arc length subtended by a central angle of  $\frac{5\pi}{6}$  radians.

- 17 The circle below has the radius and shaded sector shown. Find the area of the sector, in terms of  $\pi$ .



- 18 A sector of a circle with radius 9 has area  $27\pi$ . Find the central angle, in radians.

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### Conics: ellipses and hyperbolas

- 19 Identify the type of conic and state its center for  $\frac{(x-2)^2}{16} + \frac{(y+1)^2}{9} = 1$ .
- 20 Identify the type of conic, its center, and its asymptote slopes for  $\frac{(x+3)^2}{4} - \frac{y^2}{9} = 1$ .

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### A well-designed word problem: a Ferris wheel

- 21 This question has three parts. A circular Ferris wheel has a diameter of 40 m, with its center 22 m above the ground. Set up a coordinate system with the origin at ground level, directly below the center. (a) Write the equation of the circle traced by a passenger cabin. (b) A support cable runs from a ground anchor at (26, 0) to a point of tangency on the wheel. Find the length of this cable, to the nearest hundredth of a meter. (c) Find the height above the ground of a cabin when it is 12 m horizontally from the vertical line through the center.

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### Multiple choice — easy

**22** Write the equation of a circle with center  $(9, 12)$  and radius 12.

a.  $(x - 9)^2 + (y - 12)^2 = 12$

b.  $(x + 9)^2 + (y + 12)^2 = 144$

c.  $(x - 9)^2 + (y + 12)^2 = 144$

d.  $(x - 9)^2 + (y - 12)^2 = 144$

**23** Write the equation of a circle with center  $(4, 2)$  and radius 2.

a.  $(x + 4)^2 + (y + 2)^2 = 4$

b.  $(x - 4)^2 + (y + 2)^2 = 4$

c.  $(x - 4)^2 + (y - 2)^2 = 2$

d.  $(x - 4)^2 + (y - 2)^2 = 4$

**24** Write the equation of a circle with center  $(8, -7)$  and radius 7.

a.  $(x - 8)^2 + (y - 7)^2 = 49$

b.  $(x + 8)^2 + (y - 7)^2 = 49$

c.  $(x - 8)^2 + (y + 7)^2 = 7$

d.  $(x - 8)^2 + (y + 7)^2 = 49$

**25** Write the equation of a circle with center  $(-6, 2)$  and radius 6.

a.  $(x - 6)^2 + (y + 2)^2 = 36$

b.  $(x + 6)^2 + (y - 2)^2 = 36$

c.  $(x + 6)^2 + (y + 2)^2 = 36$

d.  $(x + 6)^2 + (y - 2)^2 = 6$

**26** Write the equation of a circle with center  $(11, -9)$  and radius 5.

a.  $(x + 11)^2 + (y - 9)^2 = 25$

b.  $(x - 11)^2 + (y + 9)^2 = 25$

c.  $(x - 11)^2 + (y + 9)^2 = 5$

d.  $(x - 11)^2 + (y - 9)^2 = 25$

**27** Write the equation of a circle with center  $(10, -2)$  and radius 10.

a.  $(x + 10)^2 + (y - 2)^2 = 100$

b.  $(x - 10)^2 + (y - 2)^2 = 100$

c.  $(x - 10)^2 + (y + 2)^2 = 10$

d.  $(x - 10)^2 + (y + 2)^2 = 100$

**28** Write the equation of a circle with center  $(0, 10)$  and radius 10.

a.  $(x)^2 + (y - 10)^2 = 102$

b.  $(x)^2 + (y - 10)^2 = 100$

c.  $(x)^2 + (y - 10)^2 = 10$

d.  $(x)^2 + (y + 10)^2 = 100$

**29** Write the equation of a circle with center  $(-11, -3)$  and radius 13.

a.  $(x + 11)^2 + (y - 3)^2 = 169$

b.  $(x + 11)^2 + (y + 3)^2 = 13$

c.  $(x + 11)^2 + (y + 3)^2 = 169$

d.  $(x - 11)^2 + (y - 3)^2 = 169$

**30** Write the equation of a circle with center  $(-5, 10)$  and radius 4.

a.  $(x - 5)^2 + (y + 10)^2 = 16$

b.  $(x + 5)^2 + (y - 10)^2 = 16$

c.  $(x + 5)^2 + (y - 10)^2 = 4$

d.  $(x + 5)^2 + (y + 10)^2 = 16$

**31** Write the equation of a circle with center  $(2, 5)$  and radius 12.

a.  $(x - 2)^2 + (y - 5)^2 = 12$

b.  $(x - 2)^2 + (y + 5)^2 = 144$

c.  $(x + 2)^2 + (y + 5)^2 = 144$

d.  $(x - 2)^2 + (y - 5)^2 = 144$

**32** Write the equation of a circle with center  $(-2, 1)$  and radius 3.

a.  $(x + 2)^2 + (y + 1)^2 = 9$

b.  $(x + 2)^2 + (y - 1)^2 = 9$

c.  $(x - 2)^2 + (y + 1)^2 = 9$

d.  $(x + 2)^2 + (y - 1)^2 = 3$

**33** Write the equation of a circle with center  $(-1, 9)$  and radius 11.

a.  $(x + 1)^2 + (y - 9)^2 = 121$

b.  $(x - 1)^2 + (y + 9)^2 = 121$

c.  $(x + 1)^2 + (y + 9)^2 = 121$

d.  $(x + 1)^2 + (y - 9)^2 = 11$

**34** Write the equation of a circle with center  $(-6, 6)$  and radius 13.

a.  $(x + 6)^2 + (y + 6)^2 = 169$

b.  $(x + 6)^2 + (y - 6)^2 = 169$

c.  $(x - 6)^2 + (y + 6)^2 = 169$

d.  $(x + 6)^2 + (y - 6)^2 = 13$

**35** Write the equation of a circle with center  $(11, -5)$  and radius 13.

a.  $(x + 11)^2 + (y - 5)^2 = 169$

b.  $(x - 11)^2 + (y + 5)^2 = 13$

c.  $(x - 11)^2 + (y - 5)^2 = 169$

d.  $(x - 11)^2 + (y + 5)^2 = 169$

**36** Write the equation of a circle with center  $(-7, 2)$  and radius 7.

a.  $(x + 7)^2 + (y - 2)^2 = 49$

b.  $(x - 7)^2 + (y + 2)^2 = 49$

c.  $(x + 7)^2 + (y + 2)^2 = 49$

d.  $(x + 7)^2 + (y - 2)^2 = 7$

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### Multiple choice — medium

**37** Find the center and radius of  $x^2 + y^2 + 2x - 2y - 62 = 0$ .

a.  $(-1, -1), 8$

b.  $(-1, 1), 8$

c.  $(1, 1), 8$

d.  $(-1, 1), 10$

38 Find the center and radius of  $x^2 + y^2 - 16x + 2y - 79 = 0$ .

- a.  $(8, 1), 12$
- b.  $(-8, -1), 12$
- c.  $(8, -1), 14$
- d.  $(8, -1), 12$

39 Find the center and radius of  $x^2 + y^2 + 12x - 6y - 99 = 0$ .

- a.  $(-6, 3), 14$
- b.  $(-6, 3), 12$
- c.  $(-6, -3), 12$
- d.  $(6, 3), 12$

40 Find the center and radius of  $x^2 + y^2 - 6x - 2y - 90 = 0$ .

- a.  $(3, 1), 12$
- b.  $(-3, 1), 10$
- c.  $(3, 1), 10$
- d.  $(3, -1), 10$

41 Find the center and radius of  $x^2 + y^2 - 16x + 12y - 21 = 0$ .

- a.  $(8, 6), 11$
- b.  $(-8, -6), 11$
- c.  $(8, -6), 11$
- d.  $(8, -6), 13$

42 Find the center and radius of  $x^2 + y^2 - 8x - 18y + 81 = 0$ .

- a.  $(-4, 9), 4$
- b.  $(4, -9), 4$
- c.  $(4, 9), 6$
- d.  $(4, 9), 4$

43 Find the center and radius of  $x^2 + y^2 - 14x - 8y + 1 = 0$ .

- a.  $(7, 4), 10$
- b.  $(7, 4), 8$
- c.  $(7, -4), 8$
- d.  $(-7, 4), 8$

44 Find the center and radius of  $x^2 + y^2 - 10x + 8y + 37 = 0$ .

- a.  $(5, 4), 2$
- b.  $(-5, -4), 2$
- c.  $(5, -4), 2$
- d.  $(5, -4), 4$

45 Find the center and radius of  $x^2 + y^2 - 14x + 2y - 50 = 0$ .

- a.  $(7, -1), 10$
- b.  $(7, -1), 12$
- c.  $(7, 1), 10$
- d.  $(-7, -1), 10$

46 Find the center and radius of  $x^2 + y^2 - 12x - 2y - 84 = 0$ .

- a.  $(6, 1), 13$
- b.  $(-6, 1), 11$
- c.  $(6, 1), 11$
- d.  $(6, -1), 11$

47 Find the center and radius of  $x^2 + y^2 + 4x + 14y - 91 = 0$ .

- a.  $(-2, -7), 12$
- b.  $(2, -7), 12$
- c.  $(-2, -7), 14$
- d.  $(-2, 7), 12$

48 Find the center and radius of  $x^2 + y^2 - 16x - 2y + 56 = 0$ .

- a.  $(8, -1), 3$
- b.  $(8, 1), 3$
- c.  $(-8, 1), 3$
- d.  $(8, 1), 5$

49 Find the center and radius of  $x^2 + y^2 + 12x - 18y + 17 = 0$ .

- a.  $(-6, 9), 12$
- b.  $(6, 9), 10$
- c.  $(-6, 9), 10$
- d.  $(-6, -9), 10$

- 50 Find the center and radius of  $x^2 + y^2 - 8y - 33 = 0$ .
- a.  $(0, -4), 7$
  - b.  $(2, 4), 7$
  - c.  $(0, 4), 9$
  - d.  $(0, 4), 7$
- 51 Find the center and radius of  $x^2 + y^2 - 2x + 4y - 116 = 0$ .
- a.  $(1, -2), 13$
  - b.  $(-1, -2), 11$
  - c.  $(1, 2), 11$
  - d.  $(1, -2), 11$

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### Multiple choice — hard

- 52 A circle has center  $O(8, 4)$  and radius 20. Point  $P(8, 33)$  lies outside the circle. Find the length of the tangent from  $P$  to the circle.
- a. 29
  - b. 23
  - c. 21
  - d. 19
- 53 A circle has center  $O(-6, 7)$  and radius 5. Point  $P(-6, 20)$  lies outside the circle. Find the length of the tangent from  $P$  to the circle.
- a. 13
  - b. 14
  - c. 12
  - d. 10

- 54 A circle has center  $O(1, 1)$  and radius 20. Point  $P(1, 30)$  lies outside the circle. Find the length of the tangent from  $P$  to the circle.
- a. 21
  - b. 23
  - c. 29
  - d. 19
- 55 A circle has center  $O(-4, -3)$  and radius 7. Point  $P(-4, 22)$  lies outside the circle. Find the length of the tangent from  $P$  to the circle.
- a. 22
  - b. 24
  - c. 25
  - d. 26
- 56 A circle has center  $O(7, 5)$  and radius 5. Point  $P(7, 18)$  lies outside the circle. Find the length of the tangent from  $P$  to the circle.
- a. 13
  - b. 12
  - c. 10
  - d. 14
- 57 A circle has radius 8. Find the area of the sector with central angle  $60^\circ$ , in terms of  $\pi$ .
- a.  $\frac{29}{3}\pi$
  - b.  $\frac{32}{3}\pi$
  - c.  $\frac{64}{3}\pi$
  - d.  $\frac{35}{3}\pi$

- 58 A circle has radius 13. Find the area of the sector with central angle  $40^\circ$ , in terms of  $\pi$ .
- a.  $\frac{338}{9}\pi$
  - b.  $\frac{178}{9}\pi$
  - c.  $\frac{160}{9}\pi$
  - d.  $\frac{169}{9}\pi$
- 59 A circle has radius 4. Find the area of the sector with central angle  $36^\circ$ , in terms of  $\pi$ .
- a.  $\frac{8}{5}\pi$
  - b.  $\frac{3}{5}\pi$
  - c.  $\frac{13}{5}\pi$
  - d.  $\frac{16}{5}\pi$
- 60 A circle has radius 12. Find the area of the sector with central angle  $150^\circ$ , in terms of  $\pi$ .
- a.  $60\pi$
  - b.  $61\pi$
  - c.  $120\pi$
  - d.  $59\pi$
- 61 A circle has radius 14. Find the area of the sector with central angle  $240^\circ$ , in terms of  $\pi$ .
- a.  $\frac{392}{3}\pi$
  - b.  $\frac{784}{3}\pi$
  - c.  $\frac{389}{3}\pi$
  - d.  $\frac{395}{3}\pi$

- 62 Find the distance between  $(1, 1)$  and  $(-1, 9)$ , to the nearest hundredth.
- a. 7.25
  - b. 8.25
  - c. 10.25
  - d. 9.25
- 63 Find the distance between  $(0, -7)$  and  $(-1, 1)$ , to the nearest hundredth.
- a. 9.06
  - b. 7.06
  - c. 8.06
  - d. 10.06
- 64 Find the distance between  $(9, -6)$  and  $(6, 9)$ , to the nearest hundredth.
- a. 14.30
  - b. 17.30
  - c. 15.30
  - d. 16.30
- 65 Find the distance between  $(5, 9)$  and  $(-2, 1)$ , to the nearest hundredth.
- a. 9.63
  - b. 12.63
  - c. 10.63
  - d. 11.63

66 Find the distance between  $(-3, 10)$  and  $(2, -3)$ , to the nearest hundredth.

a. 12.93

b. 14.93

c. 15.93

d. 13.93